

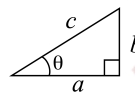
Formula sheet

Mathematics

Derivatives of powers: $\frac{dt^n}{dt} = nt^{n-1}$

Triangle geometry:

$$a^2 + b^2 = c^2; \quad a = c \cos \theta; \quad b = c \sin \theta; \quad \frac{b}{a} = \tan \theta$$



$$\sin \theta = \cos(\pi/2 - \theta); \quad \cos \theta = \sin(\pi/2 - \theta); \quad \sin \theta = \sin(\pi - \theta); \quad \cos \theta = -\cos(\pi - \theta)$$

Dot product: given vectors $\vec{A}$ and $\vec{B}$ in two dimensions (the extension to three dimensions is straightforward) with an angle θ between the directions they point in:

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}; \quad \vec{A} \cdot \vec{B} = A_x B_x + A_y B_y = |\vec{A}| |\vec{B}| \cos \theta$$

Cross Product: Given vectors $\vec{A}$ and $\vec{B}$ the magnitude of $\vec{A} \times \vec{B}$ is $|\vec{A}| |\vec{B}| \sin \theta$, and the direction is given by the right hand rule

Kinematics

In general:

$$\vec{a} = \frac{d\vec{v}}{dt}; \quad \vec{v} = \frac{d\vec{x}}{dt}; \quad (\text{And integrate to go the other way})$$

If an object is moving at a constant velocity, v , for a time, t , its position is: $x(t) = x_i + vt$.

If an object is moving with a constant *acceleration*:

$$x(t) = x_i + v_i t + \frac{a}{2} t^2; \quad v(t) = v_i + at; \quad v^2 = v_i^2 + 2a(x - x_i)$$

If an object is moving in a circle of radius R at a constant speed v (or constant *angular speed*, ω), it experiences a centripetal acceleration that point towards the center of the circle and has magnitude:

$$|\vec{a}_c| = \omega^2 R = \frac{v^2}{R}$$

If a particle is moving with respect to reference point A , and reference point A is moving with respect to reference point B , then the particle moves with respect to reference point B like:

$$\vec{v}_{P,B} = \vec{v}_{P,A} + \vec{v}_{A,B}$$

Newton's laws and forces

Newton's 2nd Law:

$$\sum \vec{F} = m\vec{a}$$

Newton's 3rd Law: If $\vec{F}_{ab}$ is a force acting by a on b , then

$$\vec{F}_{ab} = -\vec{F}_{ba}$$

Friction: For a given value of the normal force $\vec{F}_N$, the maximum force of static friction is $|f_s| = \mu_s |\vec{F}_N|$. This points in the direction which opposes the relative motion tendency of the two surfaces in contact. If the objects are sliding with respect to each other, the magnitude of the kinetic friction is $|f_k| = \mu_k |\vec{F}_N|$

Elastic force of a spring: $F_s = -kx$, where k is the spring constant, and the minus sign means the direction of F_s is always pointing back at its equilibrium position.

Work and energy

The kinetic energy of an object of mass m moving at velocity $\vec{v}$ is

$$K = \frac{1}{2}mv^2$$

The work done by a force $\vec{F}$ acting between points 1 and 2 is $W^{1 \rightarrow 2} = \int_1^2 \vec{F} \cdot d\vec{r}$

The gravitational potential energy near the Earth's surface is $U_g(y) = mgy$.

The elastic potential energy of a spring with spring constant k stretched by x away from its equilibrium position is $U_s(x) = \frac{1}{2}kx^2$.

Work - Kinetic Energy theorem for an object:

$$\sum_i W_i = \Delta(K) = \frac{m}{2}v_f^2 - \frac{m}{2}v_i^2$$

Conservation of mechanical energy says that if only conservative forces act on a system, then the sum of the kinetic and potential energies are constant:

$$E_{mech} = K_1 + U_1 = K_2 + U_2$$

Linear Momentum and center of mass

Linear momentum of a mass: $\vec{p} = m\vec{v}$. If the object's mass doesn't change with time then $\vec{F} = m\vec{a} = \frac{d\vec{p}}{dt}$.

Impulse: $\vec{I} = \Delta\vec{p} = \int_{t_1}^{t_2} \vec{F}dt$.

Collisions: perfectly elastic ones conserve both momentum and kinetic energy. Inelastic ones conserve just momentum.

Center of mass: $\vec{r}_{cm} = \frac{1}{M_{total}} \sum m_i \vec{r}_i$.

Rotational motion

Relation between rotational and linear motion quantities: arc length = $r\theta$, $v_t = r\omega$, $a = r\alpha$

Angular kinematics: $\vec{\alpha} = \frac{d\vec{\omega}}{dt}$; $\vec{\omega} = \frac{d\vec{\theta}}{dt}$, where the direction of the vectors is according to the right hand rule.

Torque about an axis: $\vec{\tau} = \vec{r} \times \vec{F}$, where $\vec{r}$ is the vector from the axis to the application point of the force.

Moment of inertia about an axis:

$$I = \sum m_i r_i^2$$

Examples of moments of inertia: $I_{CM,rod} = \frac{ML^2}{12}$, $I_{CM,disk} = \frac{MR^2}{2}$, $I_{CM,sphere} = \frac{2MR^2}{5}$

Parallel axis theorem: Let I_{CM} and I_o be the moment of inertia of an object about an axis A passing through its CM, and about another axis B parallel to axis A, respectively. D is the distance between axis A and axis B. Then:

$$I_o = I_{CM} + MD^2$$

Angular momentum about a point: $\vec{L} = \vec{r} \times \vec{p}$. For rotations about a fixed axis, this simplifies to $\vec{L} = I\vec{\omega}$
Newton's second law for rotational motion:

$$\sum \vec{\tau} = I\vec{\alpha}$$

Torque - angular momentum relation:

$$\sum \vec{\tau} = \frac{d\vec{L}}{dt}$$

Angular momentum is conserved if net torque is zero.

Rotational kinetic energy: $K = \frac{1}{2}I\omega^2$

Kinetic energy of a rigid body: $K = \frac{1}{2}Mv_{CM}^2 + \frac{1}{2}I\omega^2$

Gravitation

Newton's universal law of gravitation:

$$\vec{F}_{12} = -\frac{Gm_1m_2}{r^2}\hat{r}_{12}$$

Gravitational potential energy (when setting infinitely far as zero energy):

$$U_g(r) = -\frac{Gm_1m_2}{r}$$

Total mechanical energy of an object in an elliptical orbit with semi-major axis a (for circular orbit $a = r$):

$$E = -\frac{Gm_1m_2}{2a}$$

Kepler's laws of planetary motion: (1) Planetary orbits are ellipses and sun is at one of the focus. (2) As a planet moves in an elliptical orbit, it sweeps out equal areas in equal times. (3) There is a relationship between the period of a planet's orbit, T , and the length of its semi-major axis, a : $T^2 = a^3 \left(\frac{4\pi^2}{GM} \right)$.

Fluid mechanics

Pressure of a force acting on a surface with area A is $P = F/A$.

Pressure of a fluid at depth h :

$$P = \rho_{fluid}gh$$

Buoyant force:

$$F_B = \rho_{fluid}gV_{displaced}$$

Bernoulli's equation:

$$P + \frac{1}{2}\rho v^2 + \rho gy = constant$$

Oscillations and Waves

Simple harmonic oscillation:

$x(t) = A \cos(\omega t + \phi)$, where A is the amplitude, ω is the angular frequency, and ϕ is the initial phase.

Oscillation period T , frequency f , and angular frequency ω are related by: $T = \frac{1}{f} = \frac{2\pi}{\omega}$.

For a spring-block system, $\omega = \sqrt{\frac{k}{m}}$, and $T = 2\pi\sqrt{\frac{m}{k}}$ where k is the spring constant of the spring, and m is the mass of the block. Total mechanical energy of a spring-block simple harmonic oscillator is $E = \frac{1}{2}kA^2$.

For a simple pendulum, $\omega = \sqrt{\frac{g}{L}}$, and $T = 2\pi\sqrt{\frac{L}{g}}$ where g is the gravitational acceleration, and L is the length of the pendulum.

For a physical pendulum, $\omega = \sqrt{\frac{mgd}{I}}$ and $T = 2\pi\sqrt{\frac{I}{mgd}}$, where m is the mass of the rigid body, g is gravitational acceleration, d is the distance from the rotation axis to the center of mass, and I is the moment of inertia about the rotation axis.

Wave function of a sinusoidal wave: $y(x, t) = A \sin(kx - \omega t + \phi)$, where A is the amplitude, k is the wave number, ω is the angular frequency, and ϕ is the initial phase.

Wave number k and wave length λ are related by $k = \frac{2\pi}{\lambda}$.

The wave speed v depends only on the medium, and $v = \lambda f$.

Doppler effect: If the sound source and the observer is approaching each other, the sound frequency heard by the observer f_o is related to the frequency emitted by the source f_s by $f_o = \frac{v+v_o}{v-v_s} f_s$, where v is the sound speed in the stationary medium, v_o is the speed of the observer, and v_s is the speed of the source. If the observer and the source are moving away from each other, the equation becomes $f_o = \frac{v-v_o}{v+v_s} f_s$.

For the interference of two waves with the same amplitude A , same wave number k and same angular frequency ω , but differ by an initial phase ϕ , the resulting wave is $y = 2A \cos \frac{\phi}{2} \sin(kx - \omega t + \frac{\phi}{2})$.

Wave speed on a string $v = \sqrt{\frac{T}{\mu}}$, where T is the tension force on the string, and μ is the linear mass density of the string. Fundamental frequency of a string $f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$, where L is the length of the string.

Thermodynamics

Zeroth law of thermodynamics: If two systems are in thermal equilibrium with a third system, then they are in thermal equilibrium with each other.

First law of thermodynamics: $\Delta E_{int} = Q + W$, where $\Delta E_{int} = E_{int,f} - E_{int,i}$ is the change of the internal energy, Q and W are the heat transferred to the system and the work done on the system, respectively.

For solids and liquids:

$\Delta L = \alpha L_i \Delta T$, where $\Delta T = T_f - T_i$ is the change of temperature, α is the average coefficient of linear expansion, and $\Delta L = L_f - L_i$ is the change of length.

$\Delta V = \beta V_i \Delta T$, where $\Delta T = T_f - T_i$ is the change of temperature, β is the average coefficient of volume expansion, and $\Delta V = V_f - V_i$ is the change of volume. $\beta \approx 3\alpha$.

The heat required to raise the solid or liquid by ΔT is $Q = mc\Delta T$, where m is the mass of the solid or liquid, and c is the specific heat.

For ideal gas:

Equation of state: $PV = nRT$, where P is the pressure of the gas, V is the volume of the gas, n is the number of moles, R is universal gas constant, and T is the gas temperature.

Work done on the gas: $W = -\int_{V_i}^{V_f} P dV$.